

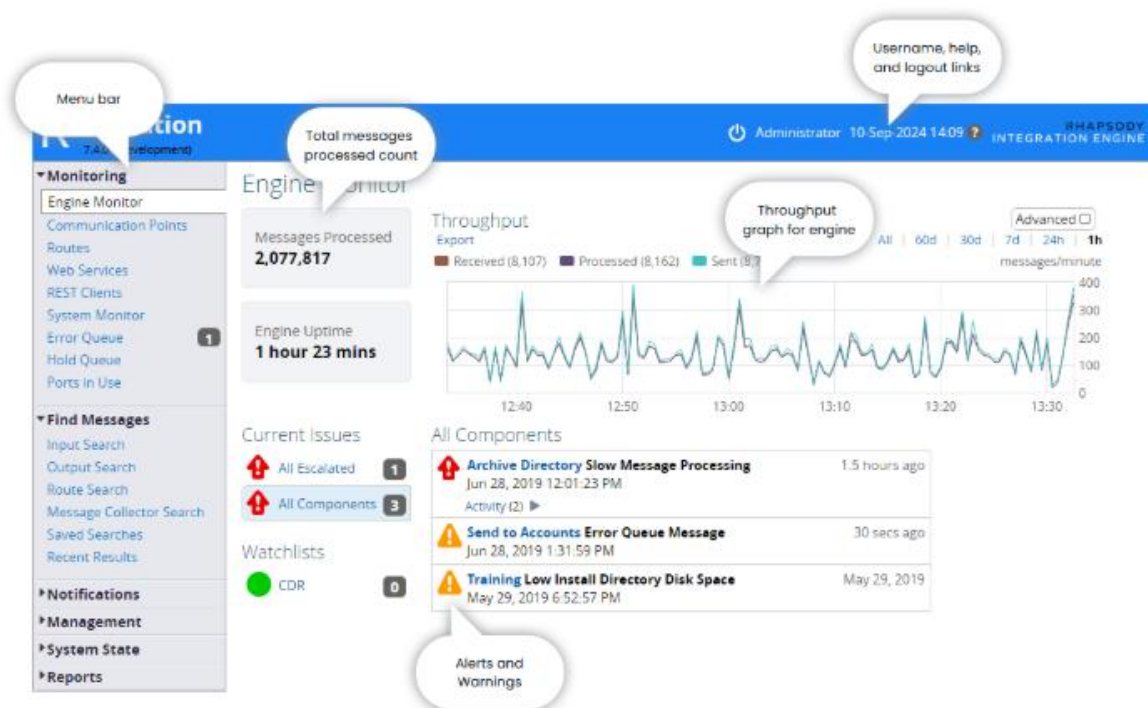
Logging in to the Management Console

Select the Management Console icon in the IDE toolbar or the Rhapsody Service icon in the Windows System Tray to launch the console and open its login screen. You can also enter the URL for the Management Console directly into a browser (by default it is `https://<rhapsody engine>:8444/rhapsody`).

The functionality you can access depends on the rights you were given by an Administrator in the IDE. All screenshots in this module show the functionality available to an Administrator. Other user groups can be configured with a reduced level of access; refer to the "Managing Your Configuration" module for configuration details.

The Management Console Interface

After you log in, the initial display will be similar to the following:



Header bar

The Header bar at the top of the screen displays the name given to the Rhapsody engine at the time of installation, along with the name of the logged in user, the current date and time and the engine's version number. You can also perform the following

View Documentation for the Management Console.

Log out of the Management Console.

Menu bar

The Menu bar lists functionality grouped by category.

- **Monitoring**
View the number of messages being processed by the engine and its individual components.
- **Find Messages**
Locate and view the content of messages processed by the engine.
- **Notifications**
Set up how the user will be notified when specified events occur.
- **Management**
Complete various administrative tasks relating to the engine, such as changing a password, message archive clearance and adding informative messages to the login screen and to the top of the console.
- **System State**
Check the log files for indications of processing problems.
- **Reports**
View performance and component statistics to identify, for example, potential bottlenecks in message processing.

You might see the following icons displayed next to the individual menu items:

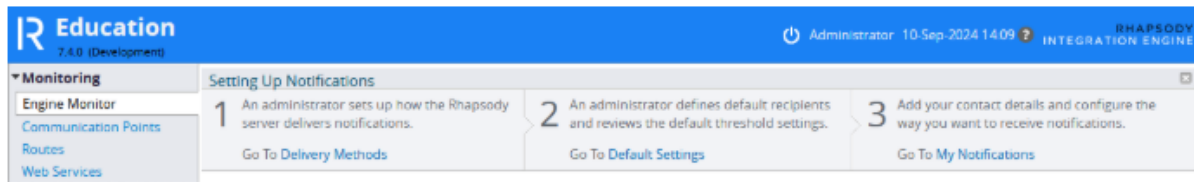
- The Message Count icon next to the Error Queue or Hold Queue menu items identifies the number of messages they currently hold.
- The Menu Alert icon is displayed next to menu items that include important tasks, such as configuring the engine's Backup settings, that have not been completed.

Notifications setup

Rhapsody notifications can be configured to alert users to issues that have occurred on the server. When notifications have not been set up, a banner will appear at the top of the Engine Monitor window with instructions.

Notifications setup

Rhapsody notifications can be configured to alert users to issues that have occurred on the server. When notifications have not been set up, a banner will appear at the top of the Engine Monitor window with instructions.



You can dismiss the banner by closing it or by completing the steps. When it has been dismissed or completed, it will not display again. However, its options will still be available on various screens under the Notifications menu.

Display window

When you select an item from the Menu bar, it opens a window to display details about that item. This window refreshes automatically every 30 seconds, but you can refresh at any time by refreshing your browser window.

Lesson 2 of 15

Engine Monitor

Rhapsody monitors the state of the components and the general health of the system. This information is presented via the Management Console. A healthy system is one in which routes and communication points are in their expected state (started or stopped) and there is minimal message queuing at any point in the configuration.

There are several indicators of interface health:

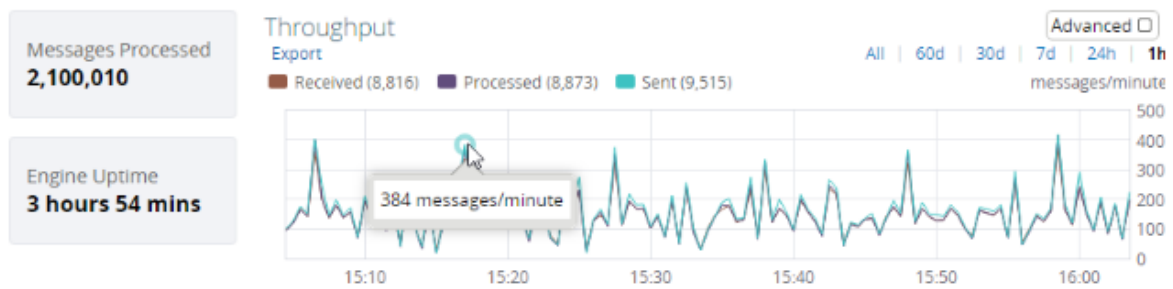
- **Issues**
Components are placed into an issue or error state if a threshold is breached and/or a trigger condition occurs.
- **Watchlists**
Individual routes, communication points and web services can be grouped into a watchlist. This allows components relating to a particular part of your solution to be monitored together for issues.
- **Error and Hold Queues**
These represent destinations for messages that generally require further scrutiny and/or action by an administrator.

- **Component Queues**
Messages will queue on a component when further processing is blocked by, for example, the non-availability of an external system.

The Engine Monitor is the first window you see when you log in. It presents an overview of the state of the engine and includes the following components:

- **Messages Processed**
The total count of messages processed by this engine.
- **Engine Uptime**
The time since the engine's last restart.
- **Throughput**
A configurable display of the engine's rate of received, processed and sent messages.

Engine Monitor



Select a different time scale by selecting the links above the graph or select and drag the mouse across a region of the graph to expand that period. Hover the mouse pointer over any point in the graph to view the message throughput at that instant in time.

Current Issues and Watchlists

The engine continually monitors the state of each component to ensure that processing is maintained within expected limits. The limits may be numeric in the form of threshold values (for example, the number of queued messages) or relate to a trigger event (for example, a communication point stopped in error). Both examples are shown in the following screenshot.

Current Issues		All Components	
 All Escalated	2	 Archive Directory Slow Message Processing	4 hours ago
 All Components	4	Jun 28, 2019 12:01:23 PM	Activity (2) ▶
Watchlists		 Send to CDR Slow Message Processing	4 hours ago
 CDR	1	Jun 28, 2019 12:01:23 PM	Activity (30) ▶
		 Send to Accounts Error Queue Message	2.5 hours ago
		Jun 28, 2019 1:31:59 PM	
		 Training Low Install Directory Disk Space	May 29, 2019
		May 29, 2019 6:52:57 PM	

Current Issues are listed on the left, with an icon indicating the highest level of severity. Selecting an issue displays its details, with links to the component(s) affected by the error.

A Watchlist is a group of one or more components whose activity is monitored for any issues. Users can be set up to receive notifications for the components in watchlists.

Lesson 3 of 15

System Monitor

The System Monitor window provides an overview of the overall system state (as opposed to the configuration state which can be viewed elsewhere), including:

- **Graphs summarizing recent message throughput, available disk space and allocated memory use.**
- **System warnings and alerts.**
- **Recently logged system events.**
- **Summary statistics.**
- **Management of system wide notifications and thresholds.**

Throughput Graph

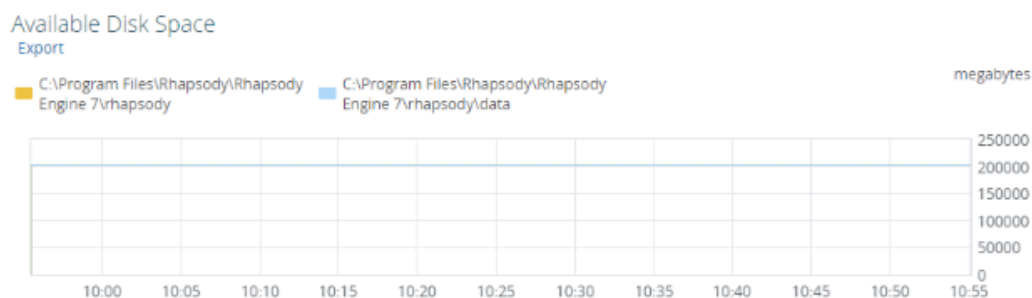
The Throughput graph presents data on Received, Processed and Sent message traffic, and is identical to the one on the Engine Monitor window. When you select Advanced, messages that were processed through the Continued Routing on Failure feature will also be reported. (This feature was discussed in the "Communication Points" module.) A configuration will frequently result in more messages being sent than received; for example, when acknowledgements are generated for each message received for processing.



Throughput over a 24-hour period varies, with peaks mid-morning and mid-afternoon, and troughs over the middle of the day. Frequently, a spike will be seen overnight due to scheduled batched processing.

Available Disk Space

The **Available Disk Space** graph shows disk usage for the installation and datastore directories. Rhapsody monitors the datastore for low disk space and will stop further message processing if insufficient space is available. This provides early warning when approaching this limit, allowing you to act prior to any issues occurring.



Rhapsody Heap Memory

The **Rhapsody Heap Memory** graph provides an overview of the heap memory used by the system. Heap memory is used dynamically by Java. The amount used often varies rapidly over time. Java regularly initiates a garbage collection to free memory no longer in use.



This graph monitors two values in its default mode:

- **Maximum heap size**
This value is defined as a parameter at engine startup.
- **Working heap**
The memory required to load the configuration and processing messages. Areas of the working heap that are above 75% of available memory are shaded red to indicate potential memory stress.

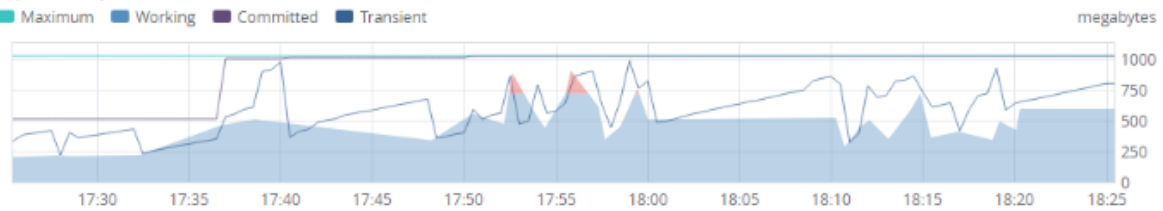
In **Advanced** mode, two additional measures are added:

Rhapsody Heap Memory

[Export](#) | [Help](#)

Advanced 

 Maximum  Working  Committed  Transient



- **Committed**

The amount of available memory guaranteed for Java use.

- **Transient**

The amount of memory in use at any time.

The **Committed** and **Transient** values can be highly variable as they depend on both the configuration and the current processing load.

The **Rhapsody Heap Memory** graph provides a snapshot of the system health by means of the **Working** memory trace. Generally, working memory allocation should lie in the lower half of the graph.

- If the working memory allocation lies between 50% and 75% of the **Maximum** memory, the system is potentially under stress. When this is the case, review the configuration and message load to determine whether this is a transient condition or whether the **Maximum Heap** allocation should be increased.
- If the **Working** memory allocation is greater than 75% of the maximum allocation, it is likely that the engine is under stress - leading ultimately to the need to restart the engine. Generally, this indicates that the maximum allocation is too low.

Issues

Warnings and alerts resulting from engine activity are listed at the top of the **Logs** panel on the lower left of the **System Monitor** window. Links are available to add a descriptive comment to each entry, or to dismiss selected entries after they have been corrected or deemed not significant.

	Low Data Directory Disk Space Jul 27, 2022 3:07:50 PM Comment Dismiss Alert Suspend Notifications Assign Activity (3) ▼	17 mins ago
	Reopened Automatically Jul 27, 2022 3:24:50 PM	6 secs ago
	Resolved Automatically Jul 27, 2022 3:22:50 PM	2 mins ago
	Changed from Warning to Alarm Jul 27, 2022 3:09:50 PM	15 mins ago

A comment can be associated with selected entries to provide further information as to the cause of the issue that generated the entry. Selecting the **Activity** link displays subsequent engine activity relating to the entry; for example, an entry identifying Low JVM Memory is likely to have an entry stating **Resolved Automatically** because of the Java engine initiating a garbage collection.

Logs

The lower panel lists entries from the Rhapsody log relating to engine activity. The list can be filtered by severity: **Trace**, **Debug**, **Info**, **Warn**, and **Error**. The **Trace** and **Debug** log levels generate a lot of information, which can be useful when troubleshooting.

All Issue Types	Info		Notifications ☐
Make a comment			
	Raised issue to alarm as there is only 5,182MB of free space available on the device mounted at '/opt/lyniate/rhapsody/rhapsody/data'. The configured disk space alarm threshold is 6,144MB.	4 mins ago	
	There is only 9,049MB of free space available on the device mounted at '/opt/lyniate/rhapsody/rhapsody/data'. The configured disk space threshold is 10,240MB.	6 mins ago	
	No old live error queue messages were detected during archive cleanup.	14 mins ago	
	Cleanup complete. There is now 11,990MB free disk space (0MB cleaned up).	14 mins ago	

Engine Information

The **Engine Information** panel is a read-only view of the engine's disk and memory usage.

Available Disk Space
7.3 GB

Available Unused Memory
605 MB

Total Process Memory
685 MB

Total Physical Memory
2.0 GB

Notifications

The upper portion of the **Notifications** panel identifies the administrators to whom warnings, alerts and escalations will be sent, along with their frequency. Links to change the notification parameters or to suspend them entirely are available next to the panel's title.

The lower panel lists the custom threshold values currently in place and includes a link to change them.

Notifications [Change](#) | [Suspend All](#)

Warnings, Alarms and Escalations

Send To support@healthyorg.internal (default)

Resend and Escalation Periods

Send Warnings Every	60 minutes
Send Alarms Every	60 minutes
Escalate After	60 minutes

Custom Thresholds [Change](#)

At All Times

Large Communication Point Queue (messages)	1000 (Warning)
	10000 (Alarm)
Large Route Queue	1000 (Warning)

Lesson 4 of 15

Exercise - Engine and System Monitor

In this exercise, we'll use the Management Console to review several engine and system statistics.

Resources

This exercise uses the RLC file available in the **. \Associate Files and Definitions\107 - Management Console** folder in the resources Zip file available at the start of the course.

Setup

Import *Exercises.rlc* into the engine to complete this exercise and the exercises that follow. The configuration generates an HL7 ADTA01 message and an HL7 ADTA31 message, which you can see in the Management Console.

Before starting this exercise, ensure the two routes and their associated communication points are running.

View the Engine Monitor

1. 1

Note the following statistics:

- Total Messages Processed
- Engine Uptime
- Total Messages Received (in the selected time period)
- Total Messages Processed (in the selected time period)
- Total Messages Sent (in the selected time period)

2. 2

Examine the Engine Monitor statistics for a time period of 1 hour. Take note of the following statistics:

- Total Messages Received
- Total Messages Sent

3. 3

Calculate the ratio of received to sent messages. This is an indicator of the number of copies of the messages that are made during processing.

View the System Monitor

1. 1

Select System Monitor from the Monitoring menu. Note the following statistics:

- Available Disk Space
- Available Unused Memory
- Total Process Memory
- Total Physical Memory

2. 2

Examine the rate at which messages are processed on the Throughput graph. Note the following statistics:

- Maximum number of messages per minute sent by the system in the last 5 minutes (approximate)
- Minimum number of messages per minute sent by the system in the last 5 minutes (approximate)
- Maximum In Use memory used by the system in the last 5 minutes (approximate)

3. 3

Examine the Working memory usage (shaded blue area) on the Rhapsody Heap Memory graph for the last 7 days.

Is the system likely to be under memory stress based on the following rules?

- **No Stress**
Working memory usage near or below 50% of the Maximum memory value.
- **Some Stress**
Working memory usage between 50% and 75% of Maximum memory value.
- **Stressed**
Working memory usage near or above 75% of the Maximum memory value.

4. Lesson 5 of 15

5. Component Lists

6. Selecting **Communication Points** or **Routes** in the **Monitoring** menu opens a window listing the currently configured communication points or routes, along with several options. The following screenshot shows the **Communication Points** window. The layout and options on the **Routes** window are similar.

7. Communication Points List

8. The following screenshot shows the **Communication Points** window with the lockers, folders and communication points created on completion of the Rhapsody Associate course.

Communication Points List

The following screenshot shows the Communication Points window with the lockers, folders and communication points created on completion of the Rhapsody Associate course.

Communication Points (31)

(1) (2) Stopped (7) Running (24) Non-Scheduled (31) academy

	COMMUNICATION POINT	TYPE	STATUS	ACTION	CONNECTI	IN QUEUE	OUT QUEU	FAILED QU	RECEIVED	SENT	FAILED	IDLE TIME
	Rhapsody Academy (16)											
	Rhapsody Associate (...)											
	REN-102 (4)											
	Message Proc...											
	1. Single M...											
	Single ...	Timer	Runni...	Stop	1	0	0	0	21,801	0	0	4 secs
	Single ...	Sink	Runni...	Stop	1	0	0	0	0	21,801	0	4 secs
	2. Batched...											
	Batch...	Timer	Runni...	Stop	1	0	0	0	10,907	0	0	4 secs
	De-bat...	Sink	Runni...	Stop	1	0	0	0	0	65,442	0	4 secs
	REN-104 (3)											
	Reload Config...											
	Delete A0...	Sink	Stopp...	Start	-	0	0	0	0	0	0	-
	Incoming ...	TCP S...	Stopp...	Start	-	0	0	0	0	0	0	-
	Outbound...	TCP C...	Stopp...	Start	-	0	0	0	0	0	0	-
	REN-105 (5)											
	TCP Client Co...											
	1. Connect...											
	Sink	Sink	Runni...	Stop	1	0	0	0	0	1	0	1 day ...

Start Stop None selected.

Reset Counters | Change Notification Thresholds | Add Notifications | Suspend Notifications | Resume Notifications | Change Log Level

Folder view

Select the **Toggle List View** icon in the top, right corner to display a compact list of all **Communication Points** (or **Routes**). This option is useful if there are a lot of them.

Communication Points (33)

(1) Stopped (24) Running (9) Non-Scheduled (33) Search by name or type

	COMMUNICATION POINT	TYPE	STATUS	ACTION	CONNECTI	IN QUEUE	OUT QUEU	FAILED QU	RECEIVED	SENT	FAILED	IDLE TIME
	Single Message Input (Rhap...	Timer	Runni...	Stop	1	0	0	0	13,643	0	0	4 secs
	Single Message Sink (Rhap...	Sink	Runni...	Stop	1	0	0	0	0	13,643	0	4 secs
	Batched Message Input (Rh...	Timer	Runni...	Stop	1	0	0	0	6,826	0	0	6 secs
	De-batched Message Sink (R...	Sink	Runni...	Stop	1	0	0	0	0	40,956	0	6 secs
	EDI Message Generator (Rh...	Timer	Stopp...	Start	-	0	0	0	0	0	0	-
	PEO Sink (Rhapsody Acade...	Sink	Runni...	Stop	1	0	0	0	0	0	0	1 hour ...

Filtering

Communication points are listed in a read-only folder hierarchy that mirrors the one in the IDE workspace. Changes to this hierarchy, such as the addition, removal or renaming of a communication point, must be completed in the IDE.

At the top of the list there are several filter buttons:

Warning

Components for which an alarm has been raised; for example, the number of retry attempts to start the component has been reached.

Stopped

Components that are not actively processing messages.

Escalated

Components for which an active alarm has been escalated to an administrator for further action.

Running

Components that are actively processing messages.

Non-Scheduled

Components that do not have an active daily or weekly schedule of operation set. Refer to the Scheduling tab on the Properties dialog for any communication point for more information.

By default, all components are listed. Use the Search field to further filter the list.

Component columns

Information related to the communication points is displayed under the following headings:

- **Type**
The communication point's type (Directory, TCP Server, etc.)
- **Status**
Whether the communication point is currently started or stopped.
- **Action**
Select to change the communication point's status.

- **Connections**
The number of connections to other components from this communication point. Connections are at the application level and may not reflect the physical connection state (particularly in the case of the TCP components).
- **In Queue**
The number of messages queued at the communication point awaiting input to a route. A non-zero count indicates that messages are delayed while waiting to be input to the route. This suggests either a high engine load or that the route is blocking due to slow processing by one or more of its components.
- **Out Queue**
The number of messages queued for output on the communication point. A non-zero count means that messages are delayed in sending to an external system by the output communication point. This typically suggests communication problems with the external system.
- **Received**
The number of messages received by the communication point.
- **Sent**
The number of messages sent by the communication point. Note that these may not necessarily be the same.
- **Idle Time**
The length of time since the communication point last processed a message. A high value may be normal for some components, though for high throughput components it may indicate that there are issues in the flow of messages to the component.

Route columns

The Routes window uses the following headings:

- **Status**
Similar to communication points.
- **Action**
Similar to communication points.
- **Waiting**
The number of messages currently waiting to enter the route.
- **Currently Processing**
The number of messages currently being processed by the route.

- **Total Processed**
The number of messages processed by this route.
- **Idle Time**
Similar to communication points.

Bulk actions

Use the checkboxes next to the components to perform actions on multiple components, such as Start or Stop, or any of the other actions at the bottom of the screen. Selecting the checkbox for a folder automatically selects its sub-folders and their components.

Lesson 6 of 15

Communication Point and Route Details

Select a communication point from the Communication Points window to open the details screen and review its layout.

Send to OneHealth (TCP Client) ▶ Stop

Healthy Org > OneHealth

Routes This Communication Point Is Used In

1

Throughput

Export

Received Sent

All | 60d | 30d | 7d | 24h | 1h
messages/minute



No Warnings or Alarms

3

Showing 0 of 0

All Issue Types	Info	Notifications
Make a comment		
☐ Communication Point Connection Failed	41 mins ago	
Show error details		
☐ Connection reset exception occurred: Failed to read from TCP stream.	42 mins ago	
☐ Communication Point Started	1 hour ago	
Jul 1, 2019 2:56:56 PM		
Comment		
☐ Communication point started	1 hour ago	
Large Queue		
Jun 28, 2019 5:24:52 PM		
Comment Activity (2)		
☐ Communication Point Stopped In Error	2 days ago	
Jun 28, 2019 5:22:51 PM		
Comment Activity (1)		
☐ Exceeded connection retry limit, shutting down communication point	2 days ago	
Show error details		
☐ Communication point stopped in error: Connection retries exceeded	2 days ago	
☐ Communication Point Connection Failed	2 days ago	
Show error details		
☐ Communication Point Connection Failed	2 days ago	
Show error details		
☐ Communication Point Connection Failed	2 days ago	
Show error details		
☐ Communication Point Connection Failed	2 days ago	
Show error details		
☐ Connection reset exception occurred: Failed to read from TCP stream.	2 days ago	
☐ Communication Point Started	Jun 28, 2019	
Jun 28, 2019 2:54:09 PM		

Messages Received / Sent / Failed

3,992 / 7,861 / 0 [View Input](#) | [View Output](#) | [View Failed](#)

4

Messages Received / Sent / Failed Since Reset

3,992 / 7,861 / 0 [Reset](#)

Mode BIDIRECTIONAL

In Queue 0

Out Queue 0

Failed Queue 0

Uptime 1 hour 4 mins

Idle Time 2 secs

Log Level [Info Change](#)

[View Error Queue Messages](#) | [View Log File](#)

Support Notes

5

The OneHealth production server is prone to unexpected outages. If this communication point cannot connect to the OneHealth server, the OneHealth support team should be contacted. Do not delete messages off the output queue as message sequencing is import in OneHealth.

[Show More](#)

Notifications [Change](#) | [Suspend All](#)

6

Warnings, Alarms and Escalations

Send To Academy User

Resend and Escalation Periods

Send Warnings 60 minutes

Every

Send Alarms Every 60 minutes

Escalate After 60 minutes

Custom Thresholds [Change](#)

7

Configuration

Startup State Startup Fourth

Start only if false

previously running

8

Issue Management

The engine continually monitors the state of components to ensure they are functioning within accepted thresholds. These thresholds are defined as a set of numeric values or states, and an issue is raised if the threshold is breached.

- Numeric values are applied to queue sizes (for example, more than ten messages are queued) or to time limits (for example, a component has been idle for more than one hour).
- State conditions are applied to component operation (for example, a component has stopped in error).

Default limits are applied globally across the engine, but the limits for any component can be set to custom values to better reflect the operating conditions for the component.

Issue Events

When a condition is breached, an **Issue Event** is triggered and a number of actions occur:

- A **Notification** message is issued as configured for the component (this might be issued by email, SMS or other communication method as configured, and is sent to the designated list of recipients).
- The issue will be reported on the component page.
- The issue will alter the state of any **Watchlist** monitoring the component. A **Watchlist** groups a set of monitored components and designates a list of recipients to be notified if an issue occurs.

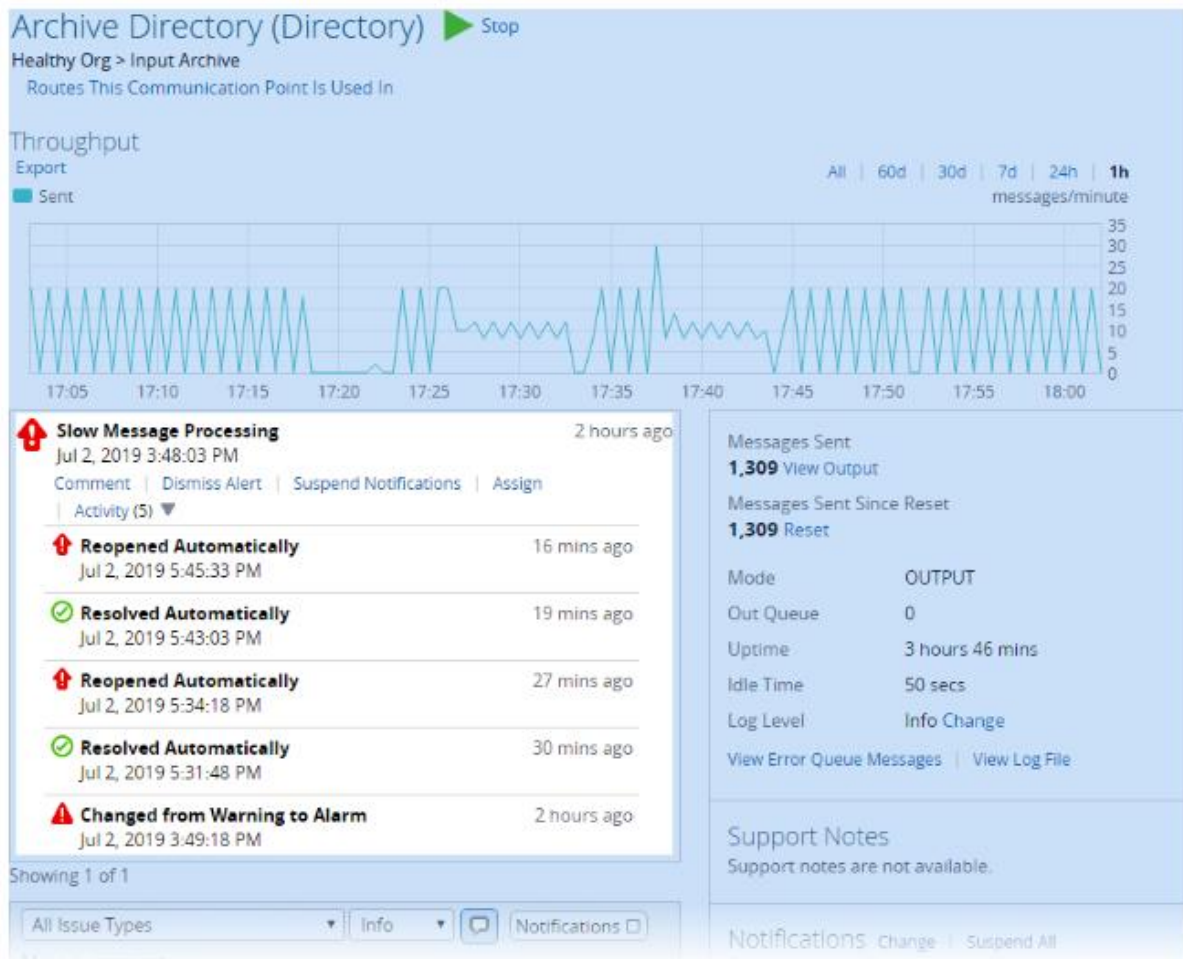
The **Issue Event** might be raised at either:

- **Warning** level, usually indicating that the issue requires attention.
- **Alert** level, usually indicating an issue requires more immediate attention; if an **Alert** level issue is raised, it can be **Escalated** to an additional recipient list if not resolved within a defined time period.

All notification messages are re-issued if the issue remains open for longer than a defined period. The **Issue Event** remains open until the issue is resolved, which might occur automatically if the engine determines that the condition no longer exists, or manually by an Administrator.

Issue Event Management

The **Event** can be managed from the component **Details** page where the **Activity Stream** for the issue is displayed for open issues.



This example shows a **Slow Message Processing** issue event which:

- Reopened automatically by the system.
- Was previously resolved automatically by the system.
- Was originally created at the Warning level before being escalated to an alarm.

Notifications and Issue management functionality ensures that open issues are visible throughout the Management Console, and are therefore difficult to ignore. They trigger regular and specific notification messages. This design ensures that one or more users are identified as responsible for the issue at all times and provides the ability to re-assign the issue as required to ensure continuity of coverage.

Lesson 8 of 15

Error and Hold Queues

The **Error** and **Hold Queues** are special queues for messages diverted from the normal processing flow because they require special attention. Messages placed on either of these queues will not affect or delay the processing of later messages, as would normally occur when FIFO constraints are in place.

Both queues display a message count alongside their entry in the Management Console's **Monitoring** menu. Selecting either queue lists the messages on that queue, sorted by date, though this can be changed by selecting a column's header. Queued message lists have a similar structure and functionality as other message lists.

Error Queue Completed Reload No saved searches

Timeframe From / To Show Last From To Search Order ↑ ↓ Hide

Search Options ▼

Communication Points Any Component ×

Routes Any Component ×

Search Reset Quick Filter

☒	TIME	COMPONENT	ROUTE	NEXT DESTINATION	ERROR DETAILS	+
☒	Jun 28, 2019 13:31:59.498	ORU messages for Accou...	Send to Accounts	Convert to Accounts JSON	Caused by: An error occur...	

1 message(s) found (1 selected)

Reinject Redirect Reprocess Delete Download Export Delete All Download All Export All

From the queue's **Message List**, you can do the following:

- Refine the search conditions to show only those messages processed within a set time span or originating from a specified communication point or route.
- Use the **Search Options** link to search for queued messages based on their input time, message type or specific content contained within the message.
- Use the plus **+** icon link to manage the displayed output columns. Use the **Quick Filter** link to specify content criteria to further filter the list.
- Select one or more messages and apply a bulk action:
 - Reinject the messages to the input communication point they were received on.
 - Redirect the messages to an alternative communication point or filter.
 - Reprocess the messages through their most recent filter (for the **Error Queue**, this is the filter at which the error occurred).
 - Download, delete or export the selected message set.

The page also provides **Delete All**, **Download All** and **Export All** operations.

Message View

Selecting a message in a **Search Results** list, or from either the **Error** or **Hold Queues** opens its **Message View**, displaying its content, associated metadata, and path through the engine. The page sections provide:

- **Message Details**
The message's unique identifier and summary detail relating to how it was selected.
- **Action Buttons**
Actions which may be applied to this message. The option depends on the search structure and the point at which the message is viewed.
- **Message Events**
A tree view of the message's processing history.
- **Message Properties**
The meta-data associated with the message at the point it was selected in the Message Events tree.
- **Message Body**
The message content associated with the message at the point it was selected in the Message Events tree.

Message View 1.1589143678208.318586

Displaying message 3 of 57 from the result set named 'Output Search' [Back to Result Set](#)

[Reinject](#)
[Redirect Copy](#)
[Redirect Edited Copy](#)
[Resend](#)

Message Events

Always Show Outputs

Last refreshed at 18:39

COMPONENT	TIMESTAMP	DESCRIPTION
Send to destinations	Jul 1, 2019 11:09:04.799	Delivered to communication point
ORU messages for CDR	Jul 1, 2019 11:09:04.799	Message dynamically routed by Dynam...
CDR	Jul 1, 2019 11:09:04.800	Received by communication point from ...
Send ORU messages to CDR	Jul 1, 2019 11:09:04.800	Placed on route
Send to CDR	Jul 1, 2019 11:09:04.801	Filtering succeeded; took 0.001 ms
	Jul 1, 2019 11:09:11.563	Delivered to communication point
	Jul 1, 2019 11:09:11.566	Registered for message tracking (Type=...
	Jul 1, 2019 11:09:11.566	Sent by communication point, total proc...
	Jul 1, 2019 11:09:11.566	Positive Acknowledgement (Type=ACKR...

Path taken by message through configuration

Message Properties

Property Name	Property Value
DefinitionName	Academy-HL7v2.4-ORU01
InputTime	1561936144797
MessageName	ORUR01
rhapsody:InputCommunicationPoint	Lab Receiver
rhapsody:InputCommunicationPointId	58
rhapsody:InputMessageType	ORUR01
rhapsody:OutputMessageType	ORUR01
router:Destination	@ORU
ValidationErrorCount	0
ValidationResult	valid

Showing 10 of 10

Message properties at current point in path

Message Body

Tree View

Flat Wire

Show Whitespace Encoding: US-ASCII

Size: 966 bytes

```
MSH|^~\&|LIS^LIS|City Hospital^City Hospital|CDR^CDR|City Hospital^City Hospital|2019080110644||ORU^R01|P1932.2019070110604|P|..
PID|1||54721-2143^80HCP|Roark^Maxine^C||18950706|P||356 Port Street ^Wellington^771^NZL^82 77V||+14 256 79012||N^Harrie
PV1|1|CH2^H2_09^80H2150|1||||Qx10-000424^Samson^Elizabeth^T^D^D^SDHA-PV^80H2150|ENC||||7||||45740^82.16.840.1.113883.3.89
ORC|RE|45758^cOrders|d184b3a205^Q5-LIS|||||Qx10-000424^Samson^Elizabeth^T^D^D^SDHA-PV^80H2150|ENC||||7||||45740^82.16.840.1.113883.3.89
OBR|1|45758^cOrders|d184b3a205^Q5-LIS|112566^Lipid Panel^OH|||20190524110904|20190524110904|20190529110644||Qx10-000424^S
OBX|1|NM|CHOL^Cholesterol^OH||5.1|mmol/L|5.1-6.1||||F|
OBX|2|NM|HDL^HDL-C^OH||1.0|mmol/L|1.0-1.5||||F|
OBX|3|NM|LDL^LDL-C^OH||4.0|mmol/L|2.6-4.1||||F|
OBX|4|NM|TG^Triglycerides^OH||1.0|mmol/L|1.7-2.2||||F|
```

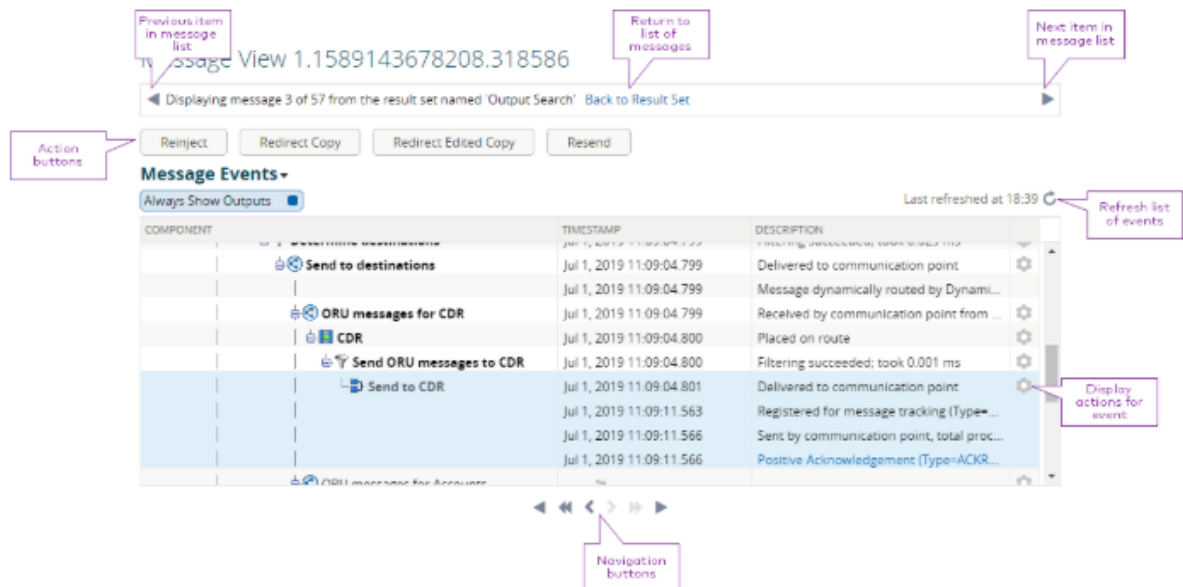
Download

Message body at current point in path

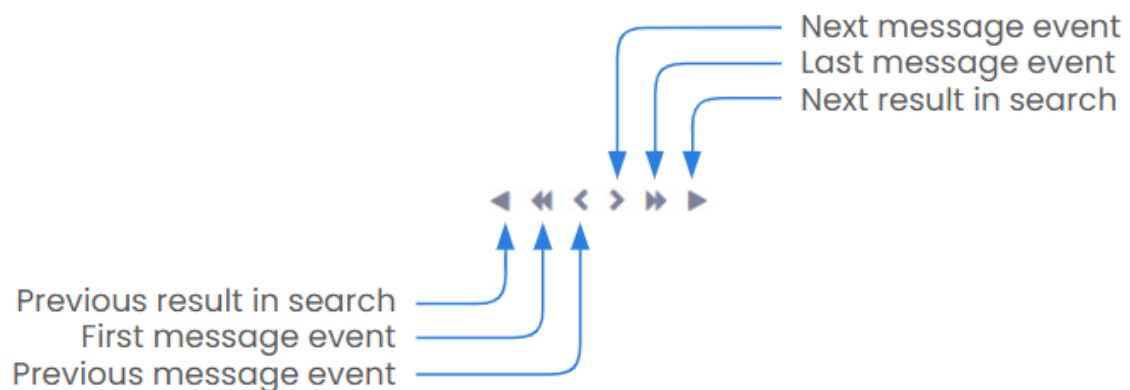
Event History

Each step in the **Message Events** tree represents the state of the message as it arrives at that step in the **Rhapsody** configuration. The light blue highlight identifies the current selection point, which determines the content of the **Message Properties** and **Message Body** panels.

In the list of events below, we show a message that has just been delivered to a communication point called **Send to CDR**.



Use the arrows at the bottom of the page to navigate through the message events.



The first message event is the communication point where the message was received.
 The last event is either the communication point where the message left Rhapsody, or the last filter the message was processed through (if the message is still being processed by Rhapsody at the point when it was viewed).

Action buttons

The **Message Action** buttons set the processing path for the current message. These options are dependent on the state of the message at the selected step. The following buttons are typical for messages in the **Error Queue**.

- **Reinject**
Place the message into the route queue as though it had just been received from a communication point.
- **Redirect Copy**
Place the message into the processing queue for any communication point or filter in the engine configuration.
- **Redirect Edited Copy**
Place an edited copy of the message into the processing queue for any component in the engine configuration.
- **Resend**
Modify the message content or properties, typically as a precursor to reprocessing. This option will only be available for messages on an Output communication point.
- **Edit**
Add new message properties and edit or delete existing properties. When the edited message is saved, a new message is created. This option will only be available for messages on the **Error** or **Hold Queues**.
- **Delete**
Remove the message from the queue. No additional processing is performed on the message. This option will only be available for messages on the **Error** or **Hold Queues**, or on live queues for route and Output communication point messages.

Message events

The **Message Events** tree is a collapsible component of the **Message View** page which presents a detailed view of each step taken by messages moving through the engine. The information includes the state of the message and its metadata as it enters the selected event.



Communication Point.



Route, also used for Extracting the route properties on entry to a route (both steps are shown in the Event Tree).



Filter, message exited via the normal output connection.



Filter, message exited via the No-Match output connection.



Filter, error occurred during processing.



Error Queue.

The **Message Events** tree provides the following details:

- **Component View**
A graphical view of the path which includes all message splits (components where multiple copies of the message are output from the component).
- **Timestamp**
The date and time to the millisecond at which the event occurred.
- **Description**
Text describing the event, including its individual processing time.

Each event also has an associated **Event Actions** icon, providing:

- **Compare In and Out**
Available only on intermediate filters, provides the ability to view the state of the message as it enters and leaves the filter.
- **Expand Descendants**
Expand the children of this event.

- **Collapse Descendants**
Collapse the tree showing the children of this event.
- **View Descendants**
View the messages associated with all children of this event.
- **View Component**
View the configuration details for this component.
- **Message Permalink**
Obtain a URL which will retrieve the message; useful for passing to other users when describing issues. It will have the form: `https://<rhapsodyserver>:8444/rhapsody/message/ViewMessage.action?id=message-1.193278439040.208897` where the 'id' is the current message's MessageID.

There is no limit to the number of events that could potentially appear in the **Message Events** tree.

Message properties

The **Message Property** section shows the properties associated with the message.

Message Properties ▾

Property Name	Property Value
DefinitionName	Academy-HL7v2.4-ORU.s3d
InputTime	1552273178327
MessageName	ORUR01
MsgCtrlId	lab2.637847b
rhapsody:InputCommunicationPoint	Incoming Lab Messages
rhapsody:InputCommunicationPointId	7110
rhapsody:InputMessageType	ORUR01
rhapsody:OutputMessageType	ORUR01

Showing 8 of 8

The example above shows the following properties:

- **DefinitionName**
The name of the message definition used to parse the message.
- **InputTime**
The timestamp for receipt of the message, expressed in milliseconds from the Java Epoch (midnight on January 1, 1970).

- **MessageName**
Derived as a consequence of parsing with a message definition.
- **MsgCtrlId**
Explicitly set during message processing.
- **rhapsody:InputCommunicationPoint**
Name of the input communication point.
- **rhapsody:InputCommunicationPointId**
The internal id for the input communication point.
- **rhapsody:InputMessageType**
Type of the input message, derived as a consequence of parsing with a message definition.
- **rhapsody:OutputMessageType**
Derived as a consequence of parsing with a message definition.

Message body

The **Message Body** is displayed in an expandable/collapsible window on the **Message View** page. The message's content can be viewed in either or both of the following ways:

Message body tree view

Tree view displays the content of HL7 messages in a hierarchical structure, provided the message body size exceeds 32KB. When viewing an HL7 message body in **Tree View**, segments and composite fields in the message can be **Expanded** or **Collapsed** by selecting their accompanying icon. This allows the message's structure to be seen, along with the data in each of its fields.



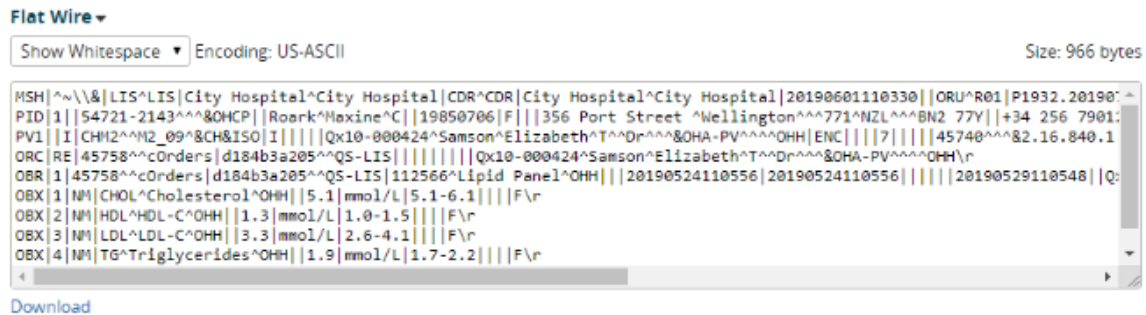
An HL7 message definition must be associated with the message for a tree view to function. If no message definition is available, or if a message is selected at the input of the **Message Events** tree, that definition has not been made available to the message. This results in the following warning:



As the message is delivered onto the route, the message definition will become available at the first filter it encounters.

Message body Flat Wire view

Flat Wire view displays the content of all messages (regardless of format) as a block of text, provided the message body size does not exceed 512KB.



The message's display format can be set to one of the following options (available dependent on the type of message):

- **Show Whitespace** (default)
Displays messages with white spaces and carriage returns.
- **Default Encoding**
Displays the message body as is. In other words, it displays the content as text without any formatting.
- **Interactive View**
When selected, selecting a field in the **Flat Wire** view expands the **Tree View** to show its corresponding location.
- **Raw Hex**
Displays the message in hexadecimal format.
- **XML**
The default display setting if the message is in XML format.
- **JSON**
Displays the message as JSON.

The current message encoding, message size and a link to **Download** the message (useful if you want to review or test the message in another application) is also displayed.

Lesson 10 of 15

Exercise - Component Monitoring

The **Communication Points** and **Routes** windows provide a high-level overview of the state of all the components in the system including:

- **State**
Stopped, Running, Retrying (when the communication point is attempting to connect to a remote system).

- **In and Out Queues**

Counts of the messages waiting to be processed or waiting to be passed on to the next component in the route. Provides a measure of the number of messages stalled on this component.

- **Sent and Received** counts

Counts of messages received by, and sent from, this component. Provides a measure of this component's activity.

- **Idle time**

The length of time since the component last processed a message.

These windows also display some important statistics. In this exercise, we'll complete several activities to help you understand how to monitor these statistics and what they mean.

Be sure you completed the setup in the previous exercise before continuing.

Exercise - Engine and System Monitor

Return to exercise and complete setup.

Filter the Display

1. 1

Select **Communication Points** in the **Monitoring** menu to view the list of communication points.

2. 2

Filter the communication points to show only components whose name includes the string rec.

Search criteria are not case sensitive, and partial strings can be specified.

Select **Stopped** or **Running** to further filter by the communication point's state.

① If you view all communication points associated with this exercise, you'll notice that message count for the **To Receiving System** communication point is twice the count for the **Generate test message** communication point. This occurs because the **1 - Create Test Messages** route creates two copies of each incoming message. As a result, the number of messages leaving the route will always be two times the number coming in.

Communication Points (23)

Stopped (8) Running (15) Non-Scheduled (23)

Filter by name: rec

	COMMUNICATION POINT	TYPE	STATUS	ACTION	CONNECT	IN QUEUE	OUT QUEUE	FAILED QU	RECEIVED	SENT	FAILED	IDLE TIME
☐	Rhapsody Academy (15)											
☐	Rhapsody Associate (15)											
☐	REN-107 (4)											
☐	Management Co...											
☐	TCP Receiver	TCP S...	Runni...	Stop	-	0	0	0	4,257	0	0	-
☐	To Receiving ...	TCP C...	Runni...	Stop	-	0	0	0	0	4,257	0	-

Component statistics

Monitor Message Throughput

- 1 On the **Communication Points** window select the **Generate test message** communication point to view its details (remember to update or remove any filter options you've applied).
- 2 Look at the **Message Throughput Graph** for the last hour. Hover the mouse pointer over the graph to view its throughput; it should be 2 messages/minute.
- 3 Return to the **Communication Points** window and select the **End of test** communication point to view its details. Confirm that its throughput averages 4 messages/minute.

View message counts

Next, we'll reset the message counts for a running communication point that is actively processing messages. This is helpful when investigating the effect of a configuration change on message flow.

1. 1

Make sure all routes and communication points provided for this exercise are started and let them run until several messages have been generated and processed.

2. 2

Manually stop the **End of test** communication point.

3. 3

Select the **End of test** communication point to view its details.

4. 4

In the **Messages Sent** panel, select the **Reset** link to reset the count of messages processed.

5. 5

Return to the **Communication Points** window and note that the message count for the **End of test** communication point displays as 0 (zero). There should also be several messages on the **Out Queue**, waiting to be processed.

6. 6

Start the **End of test** communication point. The message count should quickly increase as the **Out Queue** is emptied of unprocessed messages.

7. 7

Return to the details window. The true **Messages Sent** and **Messages Sent Since Reset** counts continue to increment as more messages are processed by the route.

Review the component log

Now we'll trigger an error on one of the communication points in the configuration. We'll use the log messages on a communication point to determine the reason why the component has failed.

⚠ Make sure all routes and communication points provided for this exercise have been started and have been running for several minutes until several messages have been generated and processed.

From the **Communication Points** window, manually stop the **TCP Receiver** communication point. After a brief period, the **To Receiving System** communication point will change to indicate that it is retrying. Then, sometime between 30 and 60 seconds, it will stop in error (the variability in the time is due to the refresh rate of the Management Console). This occurs when the retry count has been exceeded as it attempts to reconnect to the stopped server.

You should see the following:

Communication Points (23)

☐ (1)
 ☒ Stopped (8)
 ☒ Running (15)
 ☒ Non-Scheduled (23)

Search by name or type

☐	COMMUNICATION POINT	TYPE	STATUS	ACTION	CONNECT	IN QUEUE	OUT QUEU	FAILED QU	RECEIVED	SENT	FAILED	IDLE TIME
☐	Exercises (7)											
☐	Rhapsody Academy (16)											
☐	Rhapsody Associate (16)											
☐	REN-102 (4)											
☐	REN-104 (3)											
☐	REN-105 (5)											
☐	REN-107 (4)											
☐	Management Co...											
☐	▶ End of test	Sink	Runni...	Stop	1	0	0	0	0	4,151	0	9 mins
☐	▶ Generate tes...	Timer	Runni...	Stop	1	0	0	0	2,069	0	0	29 secs
☐	■ TCP Receiver	TCP S...	Stopp...	Start	-	0	0	0	4,095	0	0	-
☐	❗ To Receiving ...	TCP C...	Stopp...	Start	-	0	41	0	0	4,097	0	-

And finally, we'll view the information saved to the Rhapsody log following a communication point stopping in error.

1. Select the **To Receiving System** communication point to view its details.
2. 2

Find the log section and select the **Show error details** link for the most recent event. This will display the details of the most recent error event on this communication point.

3. 3

On the **Error Details** window, un-check **Show Stack Trace** to hide the stack trace and locate the number of retry attempts:



Depending on the resolution of your monitor and the complexity of the error, you might need to scroll to the right to see the full error details. Generally, the last **Caused by** entry on the **Error Details** window gives the root cause of the fault.

Message Search

Search Function Overview

The Management Console provides the ability to search for messages in the Datastore, including messages that are still active and have not yet completed processing. The Datastore is a single repository which has two primary purposes:

- Repository for active messages and associated metadata, allowing the engine to guarantee message delivery.
- Repository for messages that have completed processing, allowing them to be reviewed.

Searches run against the Datastore do not differentiate between the two message types. This makes it possible to search for and retrieve details about any message, regardless of its current processing state.

Find messages

Message searches are performed from the Management Console's **Find Messages** menu. The following search types are available:

- **Input Search**
Find messages based on the Input communication point they arrived on. This will also locate messages still undergoing processing.
- **Output Search**
Find messages based on the Output communication point from which they left the engine. This will only return messages whose processing has been completed.
- **Route Search**
Find messages processed or undergoing processing by one or more specified routes.
- **Message Collector Search**
Find messages waiting on a message collector. Message Collectors group messages and are discussed in Rhapsody Professional.



The **Find Messages** menu also includes the following options for managing searches:

- **Saved Searches**
A set of saved search criteria that can be recalled for later re-use.
- **Recent Results**
A list of the ten most recently run searches.

Searching for messages that are still processing

A message will only appear in a search after it has started processing through the component you are searching against.

- Queued messages waiting to be placed into a route queue will only be returned by an **Input Search**.
- Messages that are still being processed by a route or that are waiting for a stopped output communication point to be started will only be returned by an **Input Search** or a **Route Search**.

This is important to remember when there are stopped or slow processing components in your Rhapsody solution.

Datastore management

The size of the Datastore is managed to constrain the size of the archive and therefore the amount of required disk space. The store commonly holds historical messaging for a set number of days. By default, this is 28 days. This can be changed depending on site requirements and processing volumes. The Archive Cleanup management processes typically run either hourly or daily to delete old messages that have completed processing, and which are now outside the archive period.

The size of the store will be impacted by historical messages remaining on a communication point, the **Error Queue**, or **Hold Queue** for long periods. These messages will prevent the cleanup process from removing other messages from disk.

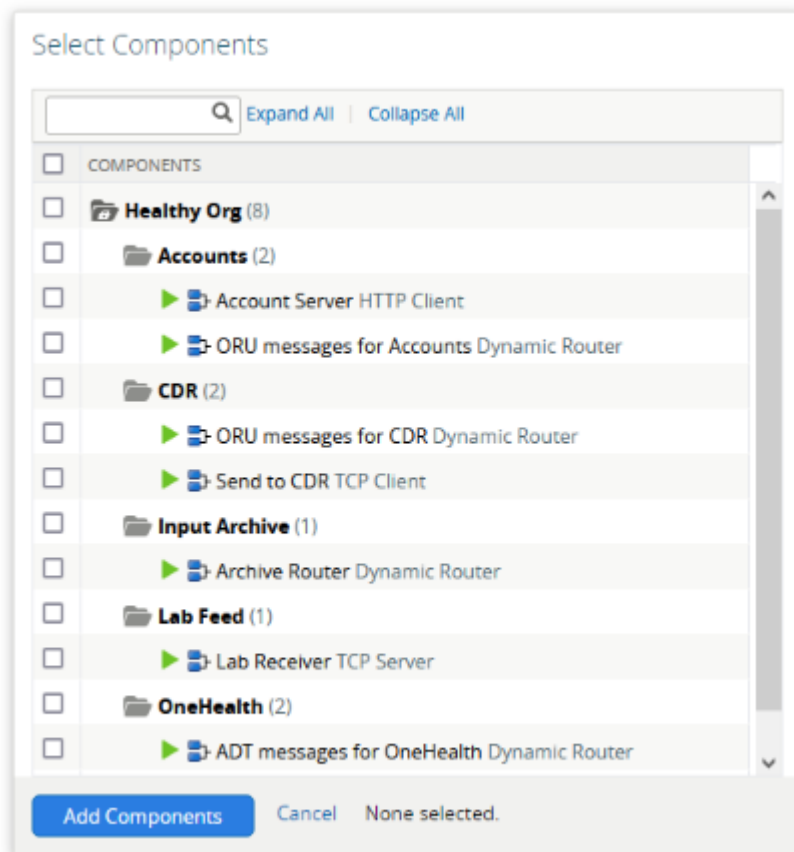
Input search

The **Input Search** finds messages processed that have been received by an Input communication point.

The screenshot shows the 'Input Search' interface. At the top, there's a header 'Input Search' and a status 'No saved searches'. Below this is a search form with several sections: 'Timeframe' with 'From / To' and 'Show Last' buttons, and 'From' and 'To' date/time pickers; 'Received By' with a search box and a dropdown menu; 'Include messages from redirected archive' with a checkbox; 'Message Contains' with a search box and radio buttons for 'Case Sensitive', 'Ignore Case', and 'Regular Expression'; and 'Search Order' with a dropdown menu. There are 'Search' and 'Reset' buttons. Below the form is a table with columns 'TIME' and 'RECEIVED BY'. The table is currently empty, and a message 'Please enter details, then press search.' is displayed at the bottom.

The default search conditions include:

- **Timeframe**
Select a time period for the search.
 - **From/To**
Specify a date range. Leaving the **To:** field blank, for example, returns all messages up to the current date and time.
 - **Show Last**
Specify the age of returned messages in either minutes, hours, days, weeks or months.
- **Search Order**
Select the order in which messages are to be sorted.
 - **Ascending**
Messages which have been in the archive longest appear at the top of the list.
 - **Descending**
The most recently processed messages appear at the top of the list.
- **Received By**
By default, all Input communication points will be applied as search criteria. To limit this number, select the folder icon and select one or more specific communication points.



- **Include messages from redirected archive**
Includes messages in the search that have been redirected to an output communication point or a filter on route.
- **Message Contains**
Specify a case-sensitive or insensitive string that returned messages must match. Alternatively, a Regular Expression can be used to search the messages. Care should be taken when using this option as it requires searching the body of each message in the search. This can result in high system load. To ensure fast searches, filter your search by specifying the communication point on which the message was received or by specifying the date range in which the message was received.

Search options

Selecting the **Search Options** link on any of the search windows displays a list of the criteria options whose display can be toggled by selecting the **Show / Hide** links.

Output Search

The **Output Search** finds messages that have completed processing and have been sent by an Output communication point. Messages that have not completed processing will not be returned by an **Output Search**.

Output Search No saved searches

Timeframe From / To Show Last From 02-Jul-2019 15:12 To Search Order 1 Search Options ▼ Hide ▼

Sent From Any Component ✕

Sent State All Succeeded Failed ✕

Received By Any Component ✕

Tracking State All messages ✕

Message Contains Case-Sensitive Ignore Case Regular Expression ✕

Search Reset

TIME	SENT FROM	SENT STATE	RECEIVED BY	TRACKING STATE
Please enter details, then press search.				

The default search conditions include the following additional options from the Input Search:

- Sent From**
 By default, all Output communication points will be included as search criteria. To limit this number, select the folder icon and select one or more specific communication points.
 - Tracking State**
 By default, all messages will be selected. To reduce the number of messages returned, use one of the **Tracking State** options; for example, **Negative Acknowledgement** or **Awaiting Reply**.
- Specifying **Tracking State** criteria allows the engine to monitor outgoing messages and match them to a response message without additional configuration. This is useful for communication points in Out->In mode where the intention is to ensure that a response is received for each message sent. Message Tracking is included in the Rhapsody Professional course.
- Message Contains**
 Specify a case-sensitive or insensitive string that returned messages must match.

Route Search

The **Route Search** finds messages that have either traversed the named route(s) or are in the process of doing so. This search is useful in locating messages processed by more than one route. By default, all routes will be included in a search. To limit this number, select the **Folder** icon to open a window where you can select individual routes

Route Search

Timeframe From / To Show Last From 02-Jul-2019 15:16 To Search Order 1 2 Search Options Hide

Routes Any Component

Received By Any Component

Message Contains Case-Sensitive Ignore Case Regular Expression

Search Reset

TIME	ROUTE	RECEIVED BY	NEXT DESTINATION
Please enter details, then press search.			

Search Results

The search results display messages that match the search criteria, along with the date and time of their processing and any other information you have chosen to include. Results are sorted in time descending order (most recent messages first). This can be changed using the **Search Order** buttons. Select a column title to sort by that column. Select it a second time to reverse the sort.

Select the **Quick Filter** link to add a row of fields to the top of the results list where you can enter search criteria. Select the plus **+** icon to modify the columns that are displayed and select the close **x** icon to remove the fields.

Output Search

Completed Reload No saved searches

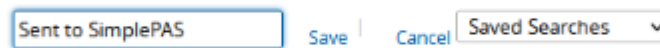
Timeframe From / To Show Last From 02-Jul-2019 15:24 To Search Order 1 2 Search Options Show

Search Reset Quick Filter

TIME	SENT FROM	SENT STATE	RECEIVED BY	TRACKING STATE
Jul 2, 2019 16:24:08.384	Message Discard	Succeeded	Sending System	
Jul 2, 2019 16:24:08.384	Message Discard	Succeeded	Sending System	
Jul 2, 2019 16:24:08.375	Message Discard	Succeeded	Sending System	
Jul 2, 2019 16:23:38.365	Message Discard	Succeeded	Sending System	
Jul 2, 2019 16:23:38.365	Message Discard	Succeeded	Sending System	
Jul 2, 2019 16:23:38.361	Message Discard	Succeeded	Sending System	

Saving and recalling search criteria

Select the save icon in the top, right corner of the panel to save the current set of criteria under a unique name. It will now be available for re-selection and reloading. The message **Saved Searches** will be displayed only when there are one or more saved searches available.



View, rename or remove your saved searches from the **Saved Searches** page, under the Management Console's **Find Messages** menu.

Quick filter

Select the **Quick Filter** link to add a row of additional criteria you can use to filter.

Action buttons

Use the checkbox next to one or more of the returned messages to perform an action. The actions behave the same as those on the **Error Queue** and **Message Details** screens. The available actions vary depending on the type of search you are viewing.

Recent results

Selecting **Recent Results** from the Management Console's **Monitoring** menu displays a list of up to ten of the user's most recent searches. The name identifies its type: **Route Search**, **Output Search** or **Input Search**. A named search, such as **Sent to SimplePAS** in the screenshot below, indicates a saved search that has been recalled and re-run.

Recent Results

Below are links to the results of your most recently run searches. Please note that Recent Results are held in memory and can be discarded by the system at any time, they will be lost after a Rhapsody restart.

Last refreshed at 15:13 

☐	Name	Message Count		Search Completed ^
☐	Sent to SimplePAS	860	View	35 secs ago
☐	Error Queue	1	View	40 secs ago
☐	Output Search	363	View	3 mins ago
☐	Input Search	16	View	3 mins ago
☒	Route Search (2)	16	View	4 mins ago
☐	Route Search	0	View	5 mins ago

Delete

1 selected

Lesson 12 of 15

Exercise - Searching for Messages

The following exercises review the basic differences between the **Input Search**, **Output Search** and **Route Search** by examining how they are used to find messages processed by the engine.

Resources

These exercises use the configuration downloaded and installed during *Exercise - Engine and System Monitor* earlier in this module. Ensure that the **1 - Create Test Messages**, **2 - Receiving System** routes and their associated communication points are running while completing this exercise.

Exercise - Engine and System Monitor

Return to exercise to complete setup.

[Go to exercise](#)

Input Search

- 1

Perform an **Input Search** on the **Generate test message** communication point on the **1 - Create Test Message** route. (Hint: The **Received By** component needs to be specified).

- 2

Select the first message from the results list to view its details on the **Message View** window.

Input Search

- 1 Perform an **Input Search** on the **Generate test message** communication point on the **1 - Create Test Message** route. (Hint: The **Received By** component needs to be specified).
- 2 Select the first message from the results list to view its details on the **Message View** window.

Model Answer

Message View 1.3094016.21857

Displaying message 1 of 120 from the result set named 'Input Search' [Back to Result Set](#)

[Reinject](#) [Redirect Copy](#) [Redirect Edited Copy](#)

Message Events

[Always Show Outputs](#)

Last refreshed at 13:47

COMPONENT	TIMESTAMP	DESCRIPTION
Generate test message	Jul 1, 2020 13:47:20.279	Received by communication point
1 - Create Test Messages	Jul 1, 2020 13:47:20.279	Placed on route
Incoming test messages	Jul 1, 2020 13:47:20.279	Filtering succeeded; took 0.002 ms
Create ADTA31 for next of kin		
Send Messages		
To Receiving System		
Send Messages		
To Receiving System		

Input message selected

Message Properties

Property Name	Property Value
InputTime	1593568040279
rhapsody.BodyEncoding	UTF8
rhapsody.InputCommunicationPoint	Generate test message
rhapsody.InputCommunicationPointId	14

Showing 4 of 4

Message Body

[Tree View](#)

[Flat Wire](#)

[Show Whitespace](#) Encoding: UTF8

Size: 499 bytes

```
MSH|^~\&|Bookings|SuburbClinic|PAS|CityHospital|20200723101022||ADT^A01|BP533155a|P|2.3\r
EVN||20200723105022\r
PID||69842-7159^MRN||Riggs^Ethan^Tony||19720423|M||585 Telescope Road^Parks^NSW^2870^AUS\r
PD1||Bough Family Clinic^1101|6x10-000998^D^Q5\r
NK1||Knox^Emilie|SIS^Sister|421 Discovery Drive^Paddys River^ACT^2620^AUS|(408) 555-3581||F|19741019\r
PV1||O|Q5_H2|R|||Qx10-000624^Hood^Leroy^B^D^Q5||MED||7|||ER44568-88111.1|||20090323101022\r
```

Output Search

- 1 Perform an **Output Search** for messages sent from the **End of test** communication point of the **2 – Receiving System** route.
- 2 Select the first message from the **Results List** to view its details on the **Message View** screen.

Model Answer

Message View 1.3128192.22090

◀ Displaying message 1 of 240 from the result set named 'Output Search (5)' [Back to Result Set](#)

Reinject

[Redirect Copy](#)

Redirect Edited Copy

[Resend](#)

Message Events ▾

Always Show Outputs ☒

Last refreshed at 13:57 

COMPONENT	TIMESTAMP	DESCRIPTION
 TCP Receiver	Jul 1, 2020 13:55:50.359	Received by communication point
 2 - Receiving System	Jul 1, 2020 13:55:50.360	Placed on route
	Jul 1, 2020 13:55:50.360	Property extraction performed
 Message Type Branch	Jul 1, 2020 13:55:50.360	Filtering succeeded; took 0.001 ms
	Jul 1, 2020 13:55:50.360	Did not match any output connector
 Send to sink	Jul 1, 2020 13:55:50.360	Filtering succeeded; took 0.001 ms
 End of test	Jul 1, 2020 13:56:54.477	Delivered to communication point
	Jul 1, 2020 13:56:54.478	Sent by communication point, total pro...

Message Properties

Property Name	Property Value
DefinitionName	Academy-HL7v2.4-ADT.s3d
InputTime	1593568550359
MessageName	ADTA01
rhapsody:InputCommunicationPoint	TCP Receiver
rhapsody:InputCommunicationPointId	17
rhapsody:InputMessageType	ADTA01
rhapsody:OutputMessageType	ADTA01

Showing 7 of 7

Message Body ▾

Tree View ▶

Flat Wire ▾

Show Whitespace Encoding: US-ASCII

Size: 499 bytes

```
MSH|^~\&|Bookings|SuburbClinic|PAS|City|Hospital|20200723101022||ADT^A01|8P533155a|P|2.3^r
EVL||20200723105022^r
PID||69842-7159^W|R||Riggs^Ethan^Tony||19720423|M||585 Telescope Road^^Parks^NSW^2870^AUS^r
PDI||Bough Family Clinic^^1101|Gx16-800998^A^D^r^Q5^r
NK1|0|Knox^Emilie|SIS^Sister^421|Discovery Drive^Paddys River^^ACT^2620^AUS|(408) 555-3581|||F|19741019^r
PVL|0|G5 H2 R||Qx16-000624^Hood^Leroy^B^D^r^Q5||R||7||ER44568-08111.1||20090523101022^r
```

Text Search

- 1 Perform a route search for messages processed in **the last hour** by the **2 - Receiving System** route and that contain the text string **Telescope Road**.
- 2 Select the first message from the results list to view its details on the **Message View** window.

Model Answer

You need to update three fields in the search to get the expected results:

- **Timeframe**
- **Routes**
- **Message Contains**

The screenshot shows the 'Route Search' interface. It includes a 'Timeframe' section with 'From / To', 'Show Last', 'Last', '1', 'hours', and a 'Search Order' dropdown. The 'Routes' section has a search box and a dropdown showing '2 - Receiving System'. The 'Received By' section has a search box and a dropdown showing 'Any Component'. The 'Message Contains' section has a search box with 'Telescope Road' and radio buttons for 'More Case' and 'Regular Expression'. A 'Search' button and a 'Reset' button are at the bottom. A 'SEARCH CRITERIA APPLIED' button is also present. A 'Quick Filter' button is at the bottom right. The results table has columns for 'TIME', 'ROUTE', 'RECEIVED BY', and 'NEXT DESTINATION'. The first three rows of the table are visible, all showing '2 - Receiving System' as the route and 'TCP Receiver' as the received by.

Route Search

Completed Reload No saved searches

Timeframe From / To Show Last Last 1 hours Search Order Search Options Hide

Routes 2 - Receiving System

Received By Any Component

Message Contains Telescope Road More Case Regular Expression

Search Reset SEARCH CRITERIA APPLIED Quick Filter

TIME	ROUTE	RECEIVED BY	NEXT DESTINATION
Jul 1, 2020 09:06:53.128	2 - Receiving System	TCP Receiver	Message Type Branch
Jul 1, 2020 09:06:23.116	2 - Receiving System	TCP Receiver	Message Type Branch
Jul 1, 2020 09:05:53.080	2 - Receiving System	TCP Receiver	Message Type Branch

Open a message to see **Telescope Road** in the message body.

Open a message to see **Telescope Road** in the message body.

Message View 1.2018176.14574

Displaying message 1 of 57 from the result set named 'Route Search (4)' [Back to Result Set](#)

Reinject

Redirect Copy

Redirect Edited Copy

Message Events ▾

Always Show Outputs ☒

Last refreshed at 09:11

COMPONENT	TIMESTAMP	DESCRIPTION	
TCP Receiver	Jul 1, 2020 09:06:53.128	Received by communication point	
2 - Receiving System	Jul 1, 2020 09:06:53.128	Placed on route	
Message Type Branch	Jul 1, 2020 09:06:53.128	Property extraction performed	
Send to sink	Jul 1, 2020 09:06:53.129	Did not match any output connector	
End of test	Jul 1, 2020 09:06:53.129	Filtering succeeded; took 0.001 ms	
	Jul 1, 2020 09:08:05.350	Delivered to communication point	
	Jul 1, 2020 09:08:05.350	Sent by communication point, total pro...	

◀ ◀ ◀ ▶ ▶ ▶ ▶ ▶

Message Properties ▾

Property Name	Property Value
InputTime	1593551213128
rhapsody:InputCommunicationPoint	TCP Receiver
rhapsody:InputCommunicationPointId	17

Showing 3 of 3

Message Body ▾

Tree View ▶

Flat Wire ▾

Show Whitespace ☐ Encoding: US-ASCII

Size: 499 bytes

```
MSH|^~\&|Bookings|SuburbClinic|PAS|CityHospital|20200723101022||ADT^A01|8P533155a|P|2.3\r
EVN||20200723105022\r
PID||69842-7159^MRN||Riggs^Ethan^Tony||19720423|M||585 Telescope Road^^Parks^NSW^2870^AUS\r
PD1||Bough Family Clinic^^1101|6x10-000998^D^Q5\r
NK1|0|Knox^Emilie|515^Sister|421 Discovery Drive^Paddys River^^ACT^2620^AUS|(408) 555-3581|F|19741019\r
PV1|0|Q5_H2|R||Qx10-000624^Hood^Leroy^B^D^Q5||MED||7|||ER44568-88111.1|20090323101022\r
```

[Download](#)

Message Processing Time

We're now going to inspect some messages in the search results looking for a message with particular characteristics.

1. 1

Search for messages sent to the **End of test** communication point and select the first message in the list.

2. 2

Navigate through the next five messages and observe the total processing time for each message reaching the output communication point. (Hint: You may have to adjust the **Description** column's width to view this information.)

3. 3

Note the maximum recorded processing time. This value will be required for the following question.

Compare Message Event Paths

Using the following screenshots, compare the event path on the **Message View** page for a message on the **1 - Create Test Messages** route with the same path for messages traversing the **2 - Receiving System** route.

Message Events ▾

Always Show Outputs ☒ Last refreshed at 10:59

COMPONENT	TIMESTAMP	DESCRIPTION	
Generat test message	Jul 1, 2020 09:15:53.423	Received by communication point	
1 - Create Test Messages	Jul 1, 2020 09:15:53.423	Placed on route	
Incoming test messages	Jul 1, 2020 09:15:53.423	Filtering succeeded; took 0.001 ms	
Create ADTA31 for next of kin			
Send Messages			
To Receiving System			
Send Messages			
To Receiving System			

Create test message route



Message Events ▾

Always Show Outputs ☒ Last refreshed at 10:59

COMPONENT	TIMESTAMP	DESCRIPTION	
Generat test message	Jul 1, 2020 09:15:53.423	Received by communication point	
1 - Create Test Messages	Jul 1, 2020 09:15:53.423	Placed on route	
Incoming test messages	Jul 1, 2020 09:15:53.423	Filtering succeeded; took 0.001 ms	
Create ADTA31 for next of kin			
Send Messages			
To Receiving System			
Send Messages			
To Receiving System			

Create test message route



In-Out Comparison – Message View

Now, we'll use the **1- Create Test Messages** route to compare the content and properties of messages before and after they have been processed by a filter.

- 1 From the Management Console's **Find Messages** menu, select **Output Search**.
- 2 Specify the **To Receiving System** communication point in the **Sent From** field.
- 3 Select the first message from the Search Results list.

Output Search Completed Reload No saved searches

Timeframe From / To Show Last From 01-Jul-2020 10:56 To Search Order 1 2

Search Options Hide

Sent From To Receiving System

Sent State All Succeeded Failed

Received By Any Component

Tracking State All messages

Message Contains Case-Sensitive Ignore Case Regular Expression

Search Reset Quick Filter

☐	TIME	SENT FROM	SENT STATE	RECEIVED BY	TRACKING STATE	+
☐	Jul 1, 2020 11:58:26.353	To Receiving System	Succeeded	Generate test message		
☐	Jul 1, 2020 11:58:26.353	To Receiving System	Succeeded	Generate test message		
☐	Jul 1, 2020 11:57:56.343	To Receiving System	Succeeded	Generate test message		
☐	Jul 1, 2020 11:57:56.342	To Receiving System	Succeeded	Generate test message		

- 4 In the **Events Tree**, select the **Create ADTA31 for next of kin** event followed by the **Event Actions** icon.

5

Select the **Compare In and Out** link to open a view of the before and after view of the message as it is processed at this event.

Message Events ▾

Always Show Outputs ☒

Last refreshed at 12:30 

COMPONENT	TIMESTAMP	DESCRIPTION	
 Generate test message	Jul 1, 2020 12:26:49.758	Received by communication point	
 1 - Create Test Messages	Jul 1, 2020 12:26:49.759	Placed on route	
 Incoming test messages	Jul 1, 2020 12:26:49.759	Filtering succeeded; took 0.001 ms	
 Create ADTA31 for next of kin	Jul 1, 2020 12:26:49.759	Filtering succeeded; took 0.184 ms	
 Send Messages	Jul 1, 2020 12:26:49.759	Filtering succeeded; took	
 To Receiving System	Jul 1, 2020 12:26:49.759	Delivered to communica	
 Send Messages	Jul 1, 2020 12:26:49.759	Sent by communication	
 To Receiving System			

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- Event Actions**
- [Compare In and Out](#)
 - [Expand Descendants](#)
 - [Collapse Descendants](#)
 - [View Descendants](#)
 - [View Component](#)
 - [Message Permalink](#)

Model Answer

The **Create ADTA31 for next of kin** filter on the **1 - Create Test Messages** route generates an **ADT^A31** message from the next of kin details in the incoming **ADT^A01** message. This can be seen in the **Message Body**, which shows the content of the incoming **ADT^A01** message next to the content of the mapped, outgoing **ADT^A31** message.

When a filter modifies message properties, this is also shown in the comparison view.

Message Properties ▾

In

Property Name	Property Value
InputTime	1593561506352
rhapsody:BodyEncoding	UTF8
rhapsody:InputCommunicationPoint	Generate test message
rhapsody:InputCommunicationPoin...	14

Showing 4 of 4

Out

Property Name	Property Value
InputTime	1593561506352
ReceivingSite	CityHospital
rhapsody:BodyEncoding	UTF8
rhapsody:InputCommunicationPoint	Generate test message
rhapsody:InputCommunicationPoin...	14

Showing 5 of 5

Message Body ▾

Tree View ▸

Flat Wire ▾

In

Show Whitespace ▾

Encoding: UTF8

Size: 499 bytes

```
MSH|^~\&|Bookings|SuburbClinic|PAS|CityHospital|20200723101022||A
EVN||20200723105022\r
PID||69842-7159^MRN||Riggs^Ethan^Tony||19720423|M||585 Telesc
PD1||Bough Family Clinic^^1101|Gx10-000998^Dr^Q5\r
NK1|0|Knox^Emilie|SIS^Sister|421 Discovery Drive^Paddys River^ACT
PV1||O|Q5_H2|R||Qx10-000624^Hood^Leroy^B^Dr^Q5||MED|||7|
```

<

>

Download

Out

Show Whitespace ▾

Encoding: UTF8

Size: 219 bytes

```
MSH|^~\&|Bookings|SuburbClinic|PAS|CityHospital|20200723101022||A
EVN||20200723105022\r
PID||999999^MRN||Knox^Emilie^||19741019|F||421 Discovery Driv
```

<

>

Download

Select the **Next Message** icon and repeat the In and Out comparison for messages processed by the **Pass original message** and **Generate Acknowledgements** filters.

Lesson 13 of 15

Error Details

Unhandled errors occurring during message processing will result in messages being placed onto the **Error Queue**, with a link to the cause of the error included in the **Message Event** tree. Such message errors will not be included in the system log without explicit configuration.

Message Events ▾

Always Show Outputs ☒

Last refreshed at 09:45 

COMPONENT	TIMESTAMP	DESCRIPTION	
 SimplePAS Input	Jul 3, 2019 09:44:54.505	Received by communication point	
 Process ADT	Jul 3, 2019 09:44:54.506	Placed on route	
 Flag source system	Jul 3, 2019 09:44:54.507	Filtering succeeded; took 0.014 ms	
 Update patient identifier	Jul 3, 2019 09:44:54.508	Filtering failed View error details	
 Error Queue	Jul 3, 2019 09:44:54.513	Delivered to the error queue	

1/20
◀ ◀◀ ◀ ▶ ▶▶ ▶

Selecting the **View Error Details** link displays a detailed error report presented as a Java Stack Trace. Note that multiple causes of an error will be represented as the error cascades, so it will be important to check all **Caused by** entries to determine the most significant one.

System and component errors occurring on the engine are logged to the **System Log**, accessed from the Management Console's **System State** menu.

System Log [Export Logs](#)

From:  

To:  

Log Level:

Number of results:

Component: 

[Search](#)

Date	Level	Component	Details
Jul 3, 2019 10:26:33.211	INFO	System	Communication point 'Receiving System' has stopped in error: Connection retries exceeded Details: IO error reconnecting
Jul 3, 2019 10:26:33.206	ERROR	Receiving System	Exceeded connection retry limit, shutting down communication point Show error details
Jul 3, 2019 10:26:33.205	WARN	Receiving System	Communication point stopped in error: Connection retries exceeded
Jul 3, 2019 10:26:23.202	INFO	Receiving System	Communication Point Connection Failed (1 more) Show error details
Jul 3, 2019 10:26:23.202	INFO	Receiving System	Communication Point Connection Failed Show error details
Jul 3, 2019 10:26:03.191	INFO	Receiving System	Communication Point Connection Failed (1 more) Show error details
Jul 3, 2019 10:26:03.191	INFO	Receiving System	Communication Point Connection Failed Show error details
Jul 3, 2019 10:26:03.190	WARN	Receiving System	Exception occurred closing connection. Show error details
Jul 3, 2019 10:25:53.181	WARN	Receiving System	Exception occurred closing connection. Show error details
Jul 3, 2019 10:25:53.181	INFO	Receiving System	Failed to flush TCP stream. Show error details
Jul 3, 2019 10:25:28.055	INFO	System	Communication point 'Sending System' has been stopped: Communication point stopped by user 'Administrator'.
Jul 3, 2019 10:25:28.055	INFO	Sending System	Communication point stopped
Jul 3, 2019 10:25:14.527	INFO	System	Communication point 'Batched Message Input' has been started: Communication point started by user 'Administrator'.
Jul 3, 2019 10:25:14.525	INFO	System	Communication point 'Detached Message IO' has been started: Communication point started by user 'Administrator'.

The following options are available for filtering the log file:

- **From/To Dates**
Set the dates and times within which the returned log entries were generated.
- **Log Level**
Select the amount of detail to include in the log.
- **Number of results**
Select the number of log entries to be displayed (100 by default).
- **Component**
Specify the component on which the log entries were generated.

Other links on this window include:

- **Export Logs**
Create a Zip file containing the log entries generated between the specified **From** and **To** dates (required). The logs can then be analyzed in a third-party application.
- **<Component Name>**
Opens the **Details** screen for the component, where the message throughput, notification and log comments, and archived messages generated by that component can be viewed.

- **Show error details**

Opens the **Java Stack Trace**, where the root cause of the error can generally be determined.

Log Levels

The operation of Rhapsody is continually monitored. Log events are reported as they occur. The central location for viewing logged content is the **System Log** in the Management Console's **System State** menu. Relevant information can also be viewed from the detail screens for individual routes, communication points and messages.

The options relating to logged information are provided by the Java log hierarchy, which determines the severity levels and verbosity of the logging. The log levels are:

- **All Logs** (default)
Display all logs.
- **Error**
Report only error events, lowest verbosity.
- **Warn**
Report warnings, errors and fatal events.
- **Info**
Include information events in the logs along with all warnings, errors and fatal alerts.
- **Debug**
Provide more detailed event information. This option is normally used when debugging a solution.
- **Trace**
Provide trace level logging as supported by the underlying code. This is normally fine-grained and very verbose and should not be enabled system-wide as it is likely to impact the performance of Rhapsody.
- **Off**
No logs are displayed.